

6 The development of antibiotic resistance in bacteria is a global concern.

A new type of drug is being developed against bacteria.

This drug has a number of effects on bacteria including making the bacteria more susceptible to antibiotics.

This drug is not an antibiotic.

(a) State the meaning of the term **antibiotic**.

(1)

(b) In 2019, 1.27 million deaths were due to antibiotic resistance.

It is estimated that this will rise to 10 million per year in 2050.

Which is the percentage increase in the number of deaths due to antibiotic resistance?

(1)

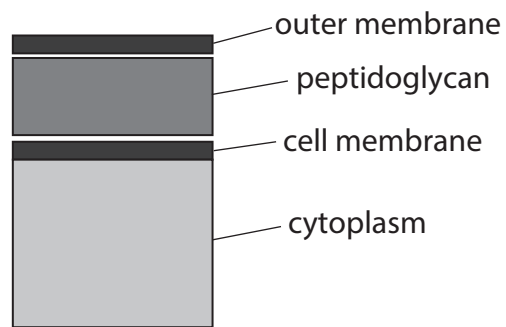
- ☐ **A** 13%
- ☐ **B** 87%
- ☐ **C** 687%
- ☐ **D** 787%

(c) Explain why it is important that these new types of drugs are developed.

(2)

- (d) The diagram shows the structure of the cell wall of a bacterial cell. It also shows the cell membrane.

The molecules in the outer membrane vary in different types of bacteria.



When bacteria were treated with the new drug, together with antibiotics, the antibiotics were effective at lower doses.

The new drug attached to the outer membrane.

Suggest why the new drug resulted in the antibiotics being effective at lower doses.

(2)

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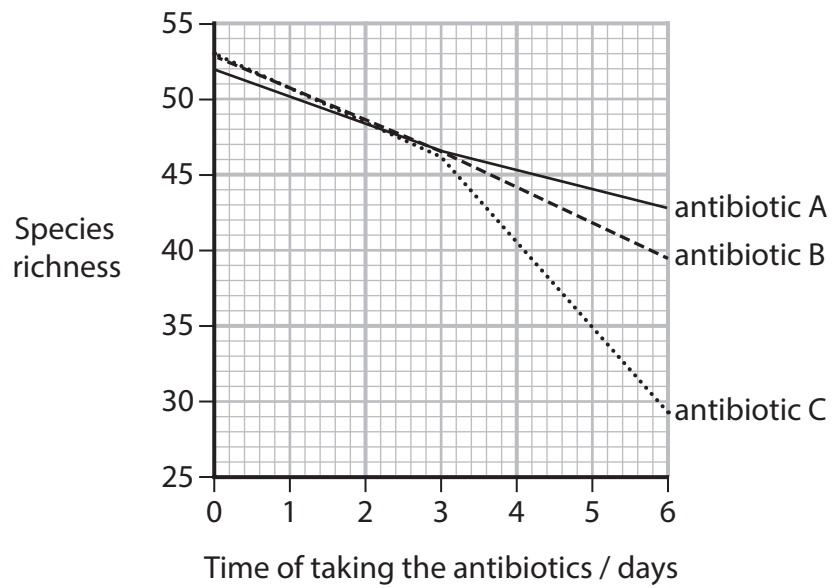
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- (e) The graph shows the effects of three antibiotics on the species richness of gut flora.



- (i) Describe **two** conclusions that can be made about the effects of these three antibiotics on the species richness of gut flora.

(2)

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(ii) Explain the advantages of adding this new drug when treating a bacterial infection with antibiotic C.

(3)

[illegible]

(Total for Question 6 = 11 marks)